JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

DEPARTMENT OF MECHANICAL ENGINEERING

DESIGN & ANALYSIS DIVISION

★ SOLVED QUESTION BANK & DERIVATIONS MANUAL ★

Kinematics of Machinery & Machine Dynamics Solved Question Bank

Course Code:	R22-MEC-2104	**Academic Scheme:**	II B.Tech I Semester (R22 Scheme)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Unit 1: Mechanisms & Inversions

Degrees of freedom, Kutzbach and Grubler criteria, Inversions of 4-bar and single/double slider-crank chains, Straight line mechanisms.

Unit 2: Velocity & Acceleration Analysis

Instantaneous center method, Kennedy's theorem, Relative velocity method, Klein's construction, Coriolis component of acceleration.

Unit 3: Cams & Followers

Types of cams and followers, Motion profiles: Uniform velocity, SHM, Uniform acceleration and Cycloidal, Cam profile synthesis.

Unit 4: Gears & Gear Trains

Law of gearing, Involute and cycloidal tooth profiles, Interference and undercutting, Epicyclic gear trains and torque calculations.

Unit 5: Gyroscopic Couples & Balancing

Precessional motion, Gyroscopic effect on aircraft, naval ships, and four-wheeled automobiles, Balancing of rotating masses in multiple planes.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Examination Scheme	70 Marks External SEE + 30 Marks Internal
Reference Textbook	Theory of Machines by S.S. Rattan / Thomas Bevan

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM